INFO 4220 – Networks II: Market Design Spring 2022 – Problem Set 1

Due date: (Friday) February 10, 11.59PM EST

Late submissions will be penalized with 1 point for every hour late.

Instructions: For each question, show all your work or you will lose some or all of the credit for that question. ***Just showing the result without showing and explaining all steps you took to find it does not count as fully correct answer.***

How to submit: Solutions must be submitted using Gradescope. For instructions on how to submit the assignment, go to this guide or watch this video.

Important: You must select the pages for each question in Gradescope. If you don't do this, you may be penalized with 3 points. If you don't know how to do this, read this guide.

Asking questions in Ed Discussions: You can use Ed Discussions to ask clarification questions about the problems. You cannot ask on Ed Discussions how to solve a particular problem, and specially, you cannot post on Ed Discussions (or anywhere else) your solution to the problem set (or a subset of it). Disclosing the solution of problems in Ed Discussions (or anywhere else) will be considered an academic integrity violation and reported accordingly. If you don't know how to solve a problem, don't ask on Ed Discussions, come to office hours! (Although, we will not solve the problem for you)

Who did you work with to solve this PS, if anyone (remember each of you must turn in their own solution, which has to be your own original work)? (0 pt)

Are you using and slip day for this PS? Remember that you have 2 for the entire semester, and they can only be used in 1-day increments. (0 pt)

Problem 1 (11 points)

After different events in the last few years, a group of people decided to spend most of their time in a virtual world. NETA, the company hosting the world, started selling virtual houses. Even though it is still a small market, it has stabilized, and it behaves like a perfectly competitive market. The following table shows the number of houses sold at different prices:

Price	Units demanded
50	900
80	840
150	700
400	200

a. If at a price of 220, NETA is willing to supply x units, and at a price of 300 NETA is willing to supply y units. Find the equations representing the demand and supply curves. Assume they are both linear functions (i.e. the Demand Curve is $Q_D(P) = A - B * P$ and the supply curve $Q_S(P) = C + D * P$. (2 pts)

We need to find the equations representing the demand and supply curve.

First, we can know the demand curve by using the given table.

$$Q_D(P) = A - B * P$$

$900 = A - B * 50 \rightarrow (1)$

$840 = A - B * 80 \rightarrow (2)$

$(1) - (2)$

$60 = 30\,B$

$B = 2, A = 1{,}000$

Therefore, $Q_D(P) = 1{,}000 - 2 * P$ (1pt)

To understand the supply curve, we only know the supply works in the price of 220 (x units) and the price of 300 (y units).

$$Q_S(P) = C + D * P$$

$x = C + D * 220 \rightarrow (3)$

$y = C + D * 300 \rightarrow (4)$

$$(4) - (3)$$

$$y - x = D * 80$$

$$D = \frac{y - x}{80}$$

If we apply D to (3), $x = C + \frac{y-x}{80} * 220$

$$C = \frac{15x - 11y}{4}$$

Therefore, $Q_S(P) = \frac{15x-11y}{4} + \frac{y-x}{80} * P$ (1pt)

b. If the equilibrium price is 400, and if NETA will not supply any houses unless the price exceeds 200. Draw the graph showing the supply and demand curves, the market equilibrium (price and quantity), the consumer surplus, and the producer surplus and their quantities. (3 pts)

Use the demand curve and supply curve found in (a),
As the equilibrium price is 400, $Q_S(400) = Q_D(400)$

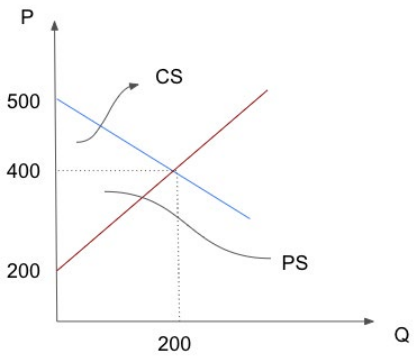
$$Q_D(400) = 1,000 - 2 * 400 = 200$$

$$Q_S(400) = 200 = \frac{15x - 11y}{4} + \frac{y - x}{80} * 400$$

$800 = 9y - 5x \rightarrow (5)$
As NETA only supply if the price exceeds 200, $Q_S(200) = 0 = \frac{15x-11y}{4} + \frac{y-x}{80} * 200$

$$0 = \frac{15x - 11y}{4} + \frac{10y - 10x}{4} = \frac{5x - y}{4}$$

$y = 5x \rightarrow (6)$
If (6) is applied to (5), $800 = 40x, x = 20, y = 100$
Therefore, $Q_S(P) = -200 + P$
When we draw the graph of both $Q_D(P)$ *and* $Q_S(P)$, it should be like,
CS (Consumer surplus) $= (500 - 400) * \frac{200}{2} = 10,000$ (0.5 pt)
PS (Producer surplus) $= (400 - 200) * \frac{200}{2} = 20,000$ (0.5 pt)



(2 pt)

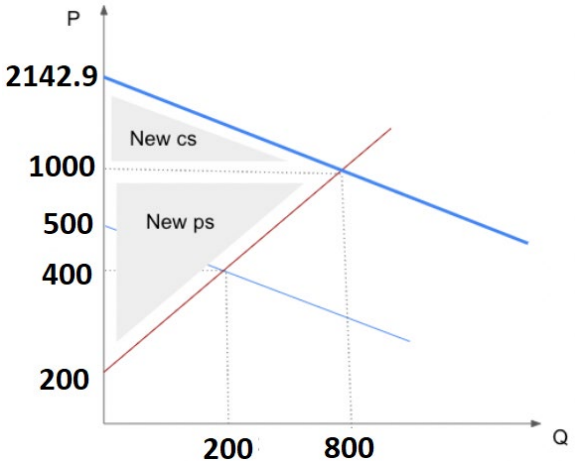
c. Due to the news that the Doomsday Clock is only 90 seconds to midnight, a wave of people started buying assets in the virtual world. If now, when the price is 0, people would demand 1500 houses, and if the price is 500, they would demand 1150 houses, what are the new demand curve and the new market equilibrium? How did the consumer and producer surplus change? (3 pts)

The demand curve should be changed according to the provided information.
Price $= 0$, House demand $= 1{,}500$
Price $=500$, House demand $= 1{,}150$
If each is calculated with $Q_S(P) = C + D * P$
new demand curve: $Q_D(P) = 1,500 - 0.7 * P$ (1pt)
For new market equilibrium, $1,500 - 0.7 * P = -200 + P$

$1.7P = 1,700$

$P = 1,000, Q = 800$ (1pt)

Therefore, new graph should be:



the consumer and producer surplus becomes

CS (Consumer surplus) $= (2,142.9 - 1,000) * \frac{800}{2} = 457,142.9$ (0.5 pt)

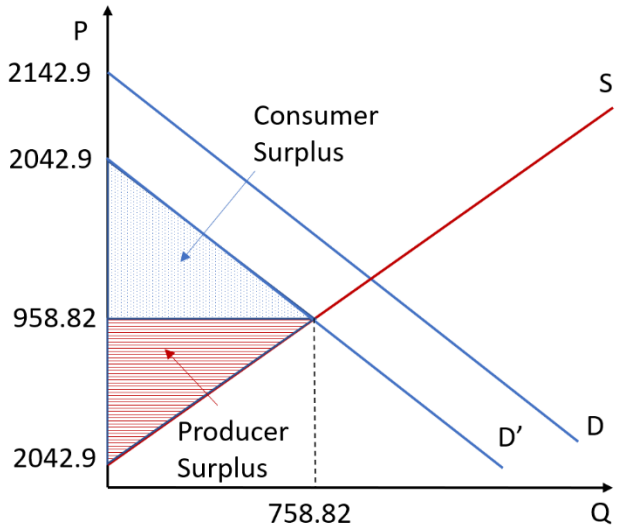
PS (Producer surplus) $= (1,000 - 200) * \frac{800}{2} = 320,000$ (0.5 pt)

d. Given the increasing amount of people in the virtual world and the emergence of hate speech, an independent government is created and starts taxing people to be able to maintain order in the virtual reality. The new virtual house tax is 100. What happened to economic welfare? Compare the changes in consumer and producer surplus? How much money will the government get from the tax? (3 pts)

As the tax is added to given price

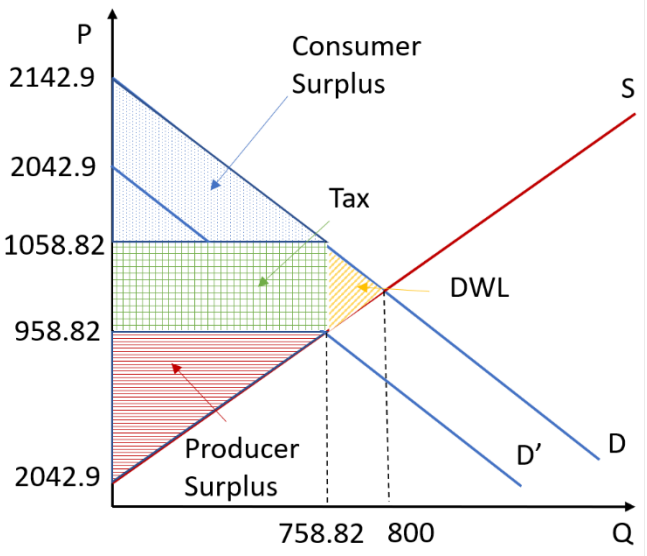
$$Q_D(P) = 1500 - 0.7 * (P + 100)$$
$$Q_D(P) = 1430 - 0.7 * P$$

Tax will shift the demand curve down.



The new equilibrium is: 1430-0.7P=-200+P => P=958.82, Q=758.82
Tax profit will be 758.82*100=7,588,2
CS=(2042.9-958.82)*758.82/2=411,295.1
PS=(958.82-200)*758.82/2=287,906.6
DWL=Total welfare part c - Total welfare part d = 2058.8

It is also possible to represent everything in just one figure. Instead of drawing the consumer surplus in the new demand curve (with the price effectively received by sellers), you can consider that the price paid by buyers is P+100 and compute the consumer surplus using the original supply curve. This has the advantage that you can also represent the tax and the deadweight loss in the same figure as follows:



Problem 2 (12 points, 3 pts each)

2.1 After winning the Pokémon World Coronation, Ash has one more challenge before retiring: Defeating you. His team comprises three Pokémon of the following types: Rock, ice, and electricity. Your team has Pokémon of the following types: Fire, water, and flying. You both know the types of Pokémon the other has, and you are both rational players. The duel will be defined in one last battle, so you must choose your Pokémon wisely. Both players will reveal the Pokémon they are playing simultaneously, and the points are awarded according to the following attack/defense matrix. Each player will get the points corresponding to an attack with the Pokémon they chose against a defense with the Pokémon their opponent chose. For example, if A chooses Fire and B chooses Ice, Player A gets 4 points and B gets 1 point. The attack/defense matrix is known to both you and Ash. Whoever gets the highest score wins.

		Defender					
Attacker		Fire	Water	Flying	Ice	Rock	Electricity
	Fire	2	1	2	4	1	2
	Water	4	2	2	2	4	1
	Flying	2	2	2	1	1	1
	Ice	1	2	4	2	4	2
	Rock	4	1	4	4	2	4
	Electricity	2	4	4	2	1	2

Iteratively delete all dominated strategies. Then determine if there is a pure strategy Nash equilibrium in this game?

Players' payoff strategies should be as below:

		Ash		
		Rock	Ice	Electricity
You	Fire	(1,4)	(4,1)	(2,2)
	Water	(4,1)	(2,2)	(1,4)
	Flying	(1,4)	(1,4)	(1,4)

First, in your case, you will never choose Flying pokemon, so we can delete using Flying pokemon. Next, in Ash case, Ash will never choose Ice pokemon, so we can delete using Ice pokemon. (1Pt)

		Ash		
		Rock	~~Ice~~	Electricity
You	Fire	(1.4)	~~(4,1)~~	(2,2)
	Water	(4,1)	~~(2,2)~~	(1,4)
	~~Flying~~	~~(1,4)~~	~~(1,4)~~	~~(1,4)~~

After deleting all dominated strategies, there is no Nash equilibrium. (2Pts)

2.2 During lecture 2, on 01/25 we played a game with the following rules: In this game, I will give you the option to collude or to defect:

- If everyone here today "colludes", everyone will get a 10 points bonus in PS1.
- If everyone but one person "colludes", the person that "defects" gets a 50 points bonus, and the rest gets 0.
- If more than one person defects, no one gets any bonus.

Find if there is a Nash Equilibrium in this game in which anyone gets a bonus? Assume that if a player's payoff is zero, there will be slightly better off if no one else gets a bonus either.

There is no Nash equilibrium in this game in which everyone gets a bonus. (1pt)

Explanation (2pts)
There are two way in which someone could get a bonus:

1) Everyone `colludes' : This is not going to be a Nash Equilibrium because if student i assumes that all other students are going to collude, then their best strategy is to defect in order to get a bonus of 50, instead of colluding ang getting a bonus of 10. This violates the N.E. conditions, so it is not a N.E.
2) Everyone but one person `colludes': It is not a Nash equilibrium because if one of the students that is defecting assumes that everyone else is going to keep their strategy (i.e. one student defects and everyone else colludes), their best strategy would be to also `defect' because we assume they will be slightly better if no one else gets a bonus either.

2.3 During lecture 3, on 01/31 we played a game with the following rules: Somehow, after grading PS1 your score and the score of your friend got lost in CANVAS (along with your submissions). Despite our best efforts, we (i.e. the teaching staff) cannot recover them. We know both of you had the exact same score and that it was in between 60 and 100, but we cannot recall the exact number. After a long meeting, we decided that we are going to ask each of you separately how much was your score. If you both say the same number, then we will give you both that score. If you say different numbers, we will know that the one of you saying the highest number is lying. Thus, we will give you both the lower score.

But, to reward the honesty of the student reporting the lower score, we will give them a +4 bonus, and to the student reporting the higher score we will give a -4 penalty.

Students were asked to enter their reported score (between 60 and 100). Payoff were determined by matching a student with a random student in the course.

Find if there is a Nash Equilibrium in this game. If there is one, demonstrate that it is a Nash Equilibrium.

The only Nash equilibrium is (60,60). (1 point)

To see this, we need to consider three cases (which can be readily seen in the table below:

- If player A reports a higher number (let's say 62) than player B (who let's say reported 61). Then player A (assuming that player B stays in 61) would change its strategy and report either the same or a a lower number than B (61 or 60), as it gets a better payoff that way
- If player A reports the same number as player B, and the number higher than 61. Both players will want to undercut the other player by 1, as by doing that they will get a higher payoff
- If both players report 60, no one has an incentive to change their strategy. If they assume that the other player will report 60, and they say a higher number, their payoff would be worse.

		friend		
		60	61	62…
student	60	(60,60)	(64, 56)	(64, 56)...
	61	(56,64)	(61,61)	(65,57)...
	62 …	(56,64)	(57,65)	(62,62)...

(2 points for the explanation)

2.4 How does the outcome of what happened in each lecture when we played the game compares with what the theory predicts (In the game from 2.2. we made two rounds, the outcome was that no one got a bonus. In the game from 2.3 the average scored entered was 94.4 and over half of the students entered a score over 97). What can you explain the similarities and differences between the theory and the outcomes of the games we played?

The basic assumptions for these games are that players are rational, know the structure of the game and its payoffs, and act strategically knowing that the others are also rational. In reality, this assumption does not always hold, especially in complex games. (0.5 points)

In game 2.2 what happened in the classroom matches what the theory predicts (as long as we assume that people prefer that no one gets a bonus if they don't get a bonus). (1 point)

In game 2.3 what happened is very far from what the theory predicts. This could be for two reasons: a) The theoretical optimal strategy in this game is not easy to figure out and requires some inductive reasoning. Many students may not have understood the game in-depth enough to be able to think of the strategy (or did not have enough time to come up with it). Alternatively, b) it may be rational to not follow the rational strategy in this game. This game is known as the traveler's dilemma and is known to produce the result we observed in class, even within people that know the game well. This indicates that people assume that others will not play the equilibrium game and instead will choose to state a high number. (1.5 points)

Problem 3 (15 points)

Alice likes to drink coffee at Gimmie Coffee. How much she enjoys her trip to Gimmie Coffee depends on the weather on the way back home, since she doesn't like going walking home during a bad storm. During bad storms, Gimmie Coffee faces higher costs, as they must pay a hazard bonus to employees if they open. On each day, Gimmie Coffee must decide if they open or not, and Alice must decide if she goes to get coffee or not. The social media manager at Gimmie Coffee is not great, so they never post if it will be open for the day or not. Thus, the only way Alice can know if it is open is by going there. Depending on whether it is Sunny or Stormy in Ithaca, their payoff functions are:

Payoff if Sunny		Alice	
		Goes	Stay
Gimmie Coffee	Close	0 , -1	2 , 0
	Open	2 , 1	3 , 0

Payoff if Stormy		Alice	
		Goes	Stays
Gimmie Coffee	Close	3 , -1	5 , 0
	Open	2 , 1	3 , 0

a. In each possible state of the world (sunny or stormy) , do the players have a dominant strategy? (2 pts)

In a sunny day, Gimmie Coffee opens, but no dominant strategies exist for Alice. (1 point)
In a stormy day, Gimmie Coffee closes, but no dominant strategies exist for Alice. (1 point)

b. Assume that Gimme has access to CoffeeForecast, a 100% accurate predictor of the weather. Suppose also that students only have access to StudentWeather, which predicts the weather incorrectly with probability p. Given that StudentWeather announces that tomorrow will be stormy, find any Bayesian Nash Equilibrium of this game indicating how the strategies used by Alice and by Gimme will depend on p. Assume players use only pure strategies. (3 pts)

(incorrect weather forecast: p)

Payoff if Sunny		Alice	
		Goes	Stay
Gimmie Coffee	Close	0 , -1	2 , 0
	Open	2 , 1	3 , 0

(correct weather forecast: 1-p)

Payoff if Stormy		Alice	
		Goes	Stays
Gimmie Coffee	Close	3 , -1	5 , 0
	Open	2 , 1	3 , 0

Notice that Gimme knows for sure if it will be sunny or not, and they have a dominant strategy. Thus, if it is sunny they will open, if it is stormy they will close. (1 point)

With that information, we can compute Alice's expected payoff for each possible action:
If Alice goes, her expected payoff is: $1 * p + (1 - p) * -1$
If Alice stays, her expected payoff: $0 * p + (1 - p) * 0$
(1 point)

Therefore when StudentWeather says stormy, she will choose to stay as long as the expected payoff of staying is greater (or equal) than the expected payoff of going.
$0 * p + (1 - p) * 0 \geq 1 * p + (1 - p) * (-1)$
$0 \geq 2p - 1$
$P \leq 1/2$ (And go otherwise)

Note that when p=1/2 it doesn't matter if Alice's stays or goes, as the expected payoff of both actions is the same
(1 point)

c. Now assume that the payoff functions are given by the tables below:

Payoff if Sunny		Alice	
		Goes	Stay
Gimme Coffee	Close	0 , -1	2 , 0
	Open	2 , 1	3 , 0

Payoff if Stormy		Alice	
		Goes	Stays
Gimme Coffee	Close	1.5 , -1	3.5 , 0
	Open	2 , 1	3, 0

In each possible state of the world (sunny or stormy), do the players have a dominant strategy? (2 pts)

If the weather is sunny, Gimme Coffee opens, while Alice has no dominant strategies.(1 point)
If the weather is stormy, no dominant strategy exists in both players. (1 point)

d. (still using the payoffs from part c) Consider the case in which the weather is stormy and Gimme Coffee assumes Alice will Go with a probability of y. Find what strategy Gimme Coffee should use as a function of y if the weather is Stormy (4 pts)

Payoff if Stormy		Alice	
		Goes (y)	Stays (1-y)
Gimme Coffee	Close	1.5 , -1	3.5 , 0
	Open	2 , 1	3, 0

#if both payoffs are considered, provide 4pts
If Gimme Coffee opens, payoff will be $2y + 3 * (1 - y) = -y + 3$
If Gimme Coffee closes, payoff will be $1.5 * y + 3.5 * (1 - y) = -2y + 3.5$
Gimme Coffee will choose to open if Stormy depending on the probability y as long as:

$$-y + 3 \geq -2y + 3.5$$
$$y \geq 0.5$$

e. (still using the payoffs from part c) StudentWeather has announced a storm. Denoting the probability that Gimme Coffee Closes if it is Stormy as x, determine the optimal strategy for Alice (this will be a function of p and x). (4 pts)

Alice will choose to go as long as her expected payoff from going is greater than the payoff of staying. The strategy for Alice is a decision that depends on what StudentWeather announces, and the probability that Gimme is open. For Gimme, we know that if it is sunny it will be open, but if it is stormy, ti will be closed with probability x. StudentWeather announced a storm, that means that it will be stormy with probability (1-p) and sunny with probability p. Alices expected payoffs are:

If it is stormy (prob 1-p),
if Alice goes, Alice's expected payoff will be $-1 * x \ + \ (1 - x) * 1$
if Alice stays, Alice's expected payoff will be $0 * x \ + \ (1 - x) * 0$
(1.5 points)

If it is sunny (prob p)
if Alice goes, Alice's payoff will be 1 (because Gimme coffee always opens)
if Alice stays, Alice's payoff will be 0
(1.5 points)

Alice goes as long as the expected payoff of going is greater or equal than the expected payoff of staying:

$$(1 - p) * (-1 * x \ + \ (1 - x) * 1) \ + \ p * 1 \geq \ (1 - p) * (0 * x \ + \ 0 * (1 - x)) \ + \ p * 0$$

$$x \leq \frac{1}{2(1-p)}$$

(1 point)

Problem 4 (12 points, 3 points each part)

There are 4 musicians and 4 singers that are trying to form pairs. The preferences of musicians and singers are as follows:

P_{m_1}	P_{m_2}	P_{m_3}	P_{m_4}
s_2	s_2	s_3	s_4
s_3	s_4	s_4	s_3
s_1	s_3	s_1	s_1
s_4	s_1	s_2	s_2

P_{S_1}	P_{S_2}	P_{S_3}	P_{S_4}
m_1	m_2	m_1	m_3
m_4	m_3	m_3	m_4
m_3	m_1	m_2	m_2
m_2	m_4	m_4	m_1

a. Find a matching using the deferred acceptance algorithm with musicians proposing. Show and explain each step of the algorithm.

Step 1.

$$m_1 -> s_2$$
$$m_2 -> s_2$$
$$m_3 -> s_3$$
$$m_4 -> s_4$$

s_2 holds offers from m_1 and m_2. They reject m_1

Step 2.

m_1 which were rejected in the previous step propose to his most preferred singers to whom he has not yet proposed.

$m_1 -> s_3$

s_3 now holds offers from m_1 and m_3. They reject m_3

Step 3.

m_3 that was rejected in the previous step can make a new offer

$m_3 -> s_4$

s_4 now holds offers from m_4 and m_3. They reject m_4

Step 4.

$m_4 -> s_3$

s_3 now holds offers from m_4 and m_1. They reject m_4

Step 5.

m_4 who was rejected in the previous step can make a new offer

$$m_4 -> s_1$$

Now each musician is matched to one singer, so there are no more rejections.

Therefore, the final matching will be

$\mu = \{(m_1, s_3), (m_2, s_2), (m_3, s_4), (m_4, s_1)\}$

#if steps are well-explained, provide 3pts

b. As the algorithm progresses, are the potential matchings for musicians and singers getting better or worse off? Explain why?

- Musicians start proposing to their most preferred singers. If they get rejected, they propose to their next most preferred singer. Therefore, they can only get worse off as the algorithm progresses. (1.5 points)
- On the other hand, singers could get better off if they receive better offers and reject their previous temporary matches. (1.5 points)

c. Is the final matching optimal for singers? Explain why or why not.

#if it explains the answer and reason for it, provide 3pts

When comparing the matrix table and matched result,
s_2 is matched to m_2 *who is their first preference*.
s_3 is matched to m_1 *who is their first preference*.
s_4 is matched to m_3 *who is their first preference*.
s_1 is matched to m_4 *who is the second option in their preferences*.

The singer optimal matching is the most preferred stable matching for singers. As singer 1 is the only one that is not matched to their most preferred musician, there cannot be another stable matching that is more preferred by singers. The only way singer 1 could get a match they like better is by making the matching of another singer worse off.

Alternatively, we know that the way to obtain the singer optimal matching is to run the deferred acceptance algorithm with singers proposing. If we do that, we obtain the same result as in part a, thus it is the singer optimal matching.

d. Is there any stable matching in which musician m_1 is matched to s_2? Show the matching or prove it is impossible (do not use any theorem from class).

No, there is not any stable matching when m_1 is matched to s_2. If m_1 is matched to s_2, (m_2, s_2) becomes a blocking pair, because they are not matched to each other, and they would prefer to be matched to each other to whatever match they can get (because m2 is s2 favorite musician, and s2 is m2 favorite musician).

(3 points for this or any other explanation that is correct and depends on the definition of stability and not in the theorems)